

Improving Probabilistic Diffusion Models with Optimal Diagonal Covariance Matching

Optimal Covariance Matching (OCM) for Efficient, High-Quality Diffusion Sampling

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Abstract & Motivation

- Diffusion sampling uses Gaussian denoisers with learned mean and fixed/learned covariance.
 - The method learns optimal diagonal covariance by matching an analytic form derived from the score.
 - They propose a new unbiased training objective: Optimal Covariance Matching (OCM).
 - The approach reduces covariance error, improving quality, recall, and likelihood with fewer NFEs.
 - It applies to DDPM, DDIM, and latent diffusion.
- Key idea: regress the optimal diagonal covariance from the score, not directly from data.

Background: DDPM and DDIM

- DDPM: forward noising $q(x_t|x_{t-1})=N(\sqrt{(1-\beta_t)}x_{t-1}, \beta_tI)$, reverse Gaussian $p_\theta(x_t|x_t)$.
- DDIM: non-Markovian formulation with $\sigma \rightarrow 0$; often uses delta covariance for $p(x_0|x_t)$.
- Heuristic covariances $(\beta, \tilde{\beta})$ suffice for large T but degrade with few steps.

 DDPM vs DDIM schematic

Why Improve Covariance?

- Small-step regimes are sensitive to covariance choice (Nichol & Dhariwal, 2021).
- Full optimal covariance via the score requires the Hessian, incurring $O(D^2)$ storage and many passes.
- Amortizing diagonal Hessian prediction retains accuracy with one extra head.

 CIFAR10 comparison vs $\tilde{\beta}/\beta$ /OCM

Analytical Covariance from Score

- For $q(\dot{x}|x)=N(\alpha x, \sigma^2 I)$, the posterior covariance $\Sigma(\dot{x})$ has a closed form derived from the score.
- The form is optimal (under KL) and depends only on $\nabla_{\dot{x}} \log q(\dot{x})$; computing it requires the Hessian $H(\dot{x})$.
- Hutchinson/Rademacher can estimate $\text{diag}(H)$ but needs many samples ($M \geq 100$).

 Analytical covariance identity

Theory

Second-order Tweedie generalization

Unbiased Optimal Covariance Matching (OCM)

- Direct regression to Rademacher averages is biased for small M .
- OCM removes the inner expectation in L2, yielding an unbiased objective.
- They report it works well in practice with $M=1$.
- Training uses a frozen pretrained score network s_θ and a lightweight $h_\phi(x_t, t)$.

 OCM objective schematic

Objective

Unbiased and sample-efficient

Applying OCM to DDPM and DDIM

- The head predicts $\text{diag}(H_t(x_t))$ to form a diagonal Σ in skip-step samplers.
- Supports DDPM $t \rightarrow t'$ and DDIM $t \rightarrow 0$ with the same mean and improved covariance.
- Only one additional forward pass per step is required.

Sampling (DDPM/DDIM)

1. Compute the mean using the pretrained score network.
2. Compute Σ via $h_\phi(x_t, t)$.
3. Sample with $x \leftarrow \mu + \Sigma^{1/2} \epsilon$ (DDPM) or draw $x_0 \sim N(\mu_0, \Sigma_0)$ (DDIM).

Application

Skip-step acceleration

Model & Training

- Score network s_θ : pretrained BaseNet (e.g., UNet/DiT).
- Hessian head h_ϕ : lightweight head sharing BaseNet features.
- Training optimizes OCM for h_ϕ with θ frozen; overhead is minimal.

 Backbone with score and diag-H heads

Training

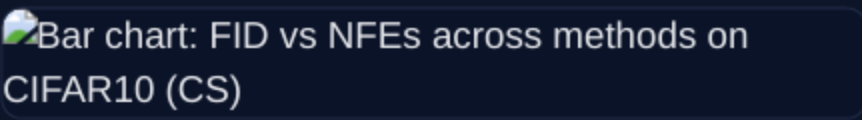
Shared backbone

Covariance Estimation Methods

Covariance Type	+#Passes	Intuition
x_t -indep. isotropic β	0	Cov of $q(x_t x_{t-1})$
x_t -indep. isotropic $\tilde{\beta}$	0	Cov of $q(x_t x_{t-1},x_0)$
Isotropic Estimation (A-DDPM)	0	Estimate from data
x_t -dep. Diag VLB (I-DDPM)	1	Learn from data
x_t -dep. Diag NS (SN-DDPM)	1	Learn from data
x_t -dep. Diag OCM (authors)	1	Learn from score
x_t -dep. Diag Est. (score)	2M	Estimate from score
x_t -dep. Diag Analytic	D	Calculate from score
x_t -dep. Full Analytic	D	Calculate from score

Results: Main Empirical Takeaways

- Across datasets, the approach achieves strong FID–NLL trade-offs at low step counts.
- It outperforms heuristic $\beta/\tilde{\beta}$ and is competitive with data-driven covariances with fewer NFEs.
- In DDIM, learning Σ_0 improves quality over delta-covariance baselines.
- In latent diffusion (DiT, ImageNet 256×256), it attains strong FID with higher Recall under CFG.



CIFAR10 (cosine schedule): FID vs NFES for β , $\tilde{\beta}$, A-DDPM, SN-/NPR-DDPM, OCM-DDPM.



CIFAR10 (cosine schedule): FID vs NLL trade-offs across covariance choices; OCM shows best balance.

Conclusion & Outlook

- OCM learns optimal diagonal covariance from the score via an unbiased loss.
- The method improves efficiency, FID, recall, and likelihood with fewer steps across models and datasets.
- Promising extensions include low-rank/block-diagonal covariances and integration with fast ODE/SDE solvers and distillation.

Better covariance enables better diffusion sampling with fewer steps.